

Advanced Programming 2025

Machine Learning versus Econometric Models for Short-Horizon Inflation Forecasting

Final Project Report

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Abstract

This project evaluates the performance of econometric and machine-learning approaches for short-horizon inflation forecasting using monthly data for eight countries over the period 1996–2019. Seasonal ARIMA models with exogenous variables (SARIMAX), Random Forest regressors, and LASSO benchmarks are compared within a rolling one-step-ahead forecasting framework. Results show that Random Forest models achieve the strongest overall forecasting performance, outperforming econometric benchmarks in several countries, while SARIMAX delivers more stable performance across the sample. Cross-country heterogeneity is substantial, with Random Forest underperforming in a small number of cases, highlighting the importance of data characteristics and late-sample dynamics. A high-dimensional Random Forest extension for Germany does not improve accuracy, illustrating that increasing feature dimensionality without careful model adaptation can degrade performance. Overall, the findings underscore the central role of persistence and seasonality in monthly inflation forecasting and demonstrate that machine-learning methods can be competitive when carefully designed and evaluated.

Keywords: data science, Python, Inflation Forecasting, Random Forest, SARIMAX, Time-Series Analysis

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1 Introduction

Accurate inflation forecasting is a central task in macroeconomic analysis, with direct relevance for monetary policy, wage negotiations, and real-time economic monitoring. Traditionally, inflation forecasts rely on econometric time-series models that explicitly exploit temporal dependence and seasonal patterns. In recent years, machine learning methods have gained popularity in applied macroeconomics due to their flexibility and ability to handle high-dimensional information sets. However, empirical evidence on whether such methods systematically outperform traditional benchmarks in small-sample macroeconomic settings remains mixed.

This project addresses the question of whether machine learning models can improve CPI inflation forecasts relative to an econometric benchmark when applied to monthly macroeconomic data with limited sample sizes.

The primary objective is to compare three forecasting approaches: SARIMAX, Random Forest regressor, and LASSO regression. All models are evaluated using a rolling forecasting design and harmonized performance metrics. Beyond forecast accuracy, the project aims to demonstrate sound programming practices, including modular system design, robust data preprocessing and reproducible experimentation. A secondary objective is to assess whether expanding the information set through a high-dimensional machine learning extension yields measurable forecast improvements.

2 Literature Review

The literature on inflation forecasting can broadly be divided into two strands: traditional econometric time-series models and more recent machine learning approaches. This project draws on both strands, combining insights from classical ARIMA-based forecasting and modern supervised learning methods.

Econometric time-series approaches. Autoregressive Integrated Moving Average (ARIMA) models and their extensions remain a standard benchmark for inflation forecasting. Early contributions emphasize the ability of ARIMA models to capture persistence and seasonality in inflation dynamics, particularly at monthly frequencies. (Junttila 2001) shows that rolling and recursive ARIMA specifications can deliver competitive inflation forecasts. These findings motivate the use of rolling estimation schemes rather than fixed-sample models in applied forecasting settings.

More generally, the econometric literature highlights the sensitivity of inflation forecasts to structural instability and regime shifts. Ignoring such features can degrade forecast accuracy, reinforcing the importance of recursive or rolling estimation designs in empirical inflation forecasting exercises (Junttila 2001).

Machine learning methods in forecasting. More recently, machine learning methods have been proposed as flexible alternatives to traditional econometric models. Regularization-based linear models such as LASSO have been widely adopted in macroeconomic forecasting to handle potentially high-dimensional predictor sets by enforcing sparsity. In this context, LASSO is often interpreted as a compromise between fully unrestricted regressions and parsimonious benchmark models, retaining only predictors that contribute meaningfully to forecast performance.

Ensemble methods, and Random Forests in particular, have attracted growing attention due to their ability to capture nonlinearities and interactions without requiring explicit functional form assumptions. While Random Forests are widely used in data-rich forecasting environments, their application to macroeconomic time series poses challenges related to small sample sizes, temporal dependence, and overfitting. (Beck, Kozbur, and Wolf 2024) show that Random Forest based inflation forecasts can outperform traditional benchmarks when very large information sets are available, but emphasize that careful regularization and feature selection are crucial for

achieving such gains.

Related work also suggests that the benefits of machine learning methods are context-dependent. (Stock and Watson 2016) demonstrate that even in linear frameworks, carefully selected predictors and parsimonious specifications can remain highly competitive, particularly in environments with limited signal-to-noise ratios. These findings align with a broader empirical consensus that increased model flexibility does not automatically translate into superior forecast accuracy in macroeconomic applications.

3 Methodology

3.1 Data Description

The analysis uses monthly macroeconomic data from two primary sources. Consumer Price Index (CPI) data, obtained from the OECD data portal (OECD 2025) the target variable. Unemployment rates and import price indices are sourced from Trading Economics (Trading Economics 2025).

The final sample covers eight OECD economies (Belgium, Germany, Israel, Korea, Latvia, Lithuania, Norway, and Switzerland) over the period January 1996 to December 2019. Country selection is based on data availability, as all three core variables require sufficiently long and consistent time series. The sample is truncated at the end of 2019 to exclude the COVID-19 period, which introduces large structural breaks that are difficult to accommodate within time series frameworks such as SARIMAX. Monthly frequency is chosen to maximize sample size, which is important for training models on the data.

The target variable is year-on-year (YoY) CPI inflation, constructed as a 12-month log difference of the CPI level series. CPI data are indexed to a common base year (2015 = 100), which serves as the reference point for rescaling and transforming all other variables.

Two exogenous variables are included in the baseline specification. (i) Unemployment is used in levels as a reduced-form indicator of cyclical economic slack, consistent with Phillips-curve type relationships. (ii) Import price indices are included to capture external cost pressures and pass-through effects; these series are rebased to the CPI base year and transformed into YoY log differences.

Data availability varies across countries, as some exogenous series begin later than CPI data, resulting in shorter effective training samples after transformation. This uneven coverage may adversely affect forecast performance for a subset of countries.

A high-dimensional Random Forest extension is implemented for Germany only. This extension incorporates a broader set of macroeconomic indicators covering prices, real activity, financial conditions, and external balances.

3.2 Approach

The project implements a controlled comparison between a seasonal econometric benchmark and two machine learning baselines for short-horizon inflation forecasting. The benchmark model is a seasonal ARIMA specification with exogenous regressors (SARIMAX), which explicitly captures monthly seasonality. In addition, a non-seasonal ARIMAX pipeline is implemented in the codebase as an intermediate benchmark to validate the treatment of exogenous variables; however, empirical results favor the seasonal specification, and SARIMAX is therefore retained as the reference model. The machine learning approaches consist of a Random Forest regressor, designed to capture nonlinearities and interactions, and a LASSO regression, used as a sparse linear benchmark.

The evaluation target is year-on-year (YoY) CPI inflation, constructed as a 12-month log

difference of the CPI level series ¹. Because SARIMAX models are estimated on CPI levels in order to explicitly capture seasonality, level forecasts are subsequently transformed into YoY inflation to ensure comparability with machine-learning outputs.

To prevent look-ahead bias, all explanatory variables enter the models exclusively in lagged form. A month-of-year indicator is additionally included to capture seasonal patterns. This lag structure is specified *ex ante* and applied uniformly (more detail in 4.1).

For the SARIMAX specification, exogenous regressors also enter only with lags; however, unlike machine-learning models, the number of lags is selected prior to evaluation using a dedicated lag-selection procedure based on information criteria. All transformations, lag construction, and trimming induced by missing observations are handled centrally within the preprocessing pipeline, ensuring consistent sample alignment.

All models are estimated within a rolling one-step-ahead forecasting framework with an expanding training window. At each forecast origin, models are re-estimated using only information available up to that date and then used to predict the next monthly observation. This procedure is applied uniformly across econometric and machine-learning models, ensuring that differences in performance reflect modeling choices rather than differences in evaluation design. The rolling framework closely mimics a real-time forecasting exercise and prevents information leakage or ex-post tuning from contaminating the results.

Forecast accuracy is assessed on year-on-year inflation using two standard loss functions for continuous outcomes: the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE). RMSE penalizes large forecast errors more strongly, while MAE provides a linear measure of average forecast error that is more robust to outliers. All metrics are computed over a common out-of-sample period and on harmonized forecast–realization pairs.

3.3 Implementation

The project is implemented in Python (PyCharm). Data manipulation and preprocessing rely on `pandas` and `numpy`, which are used to align multi country monthly time series, construct lagged features, and apply vectorized transformations. Econometric models are implemented using `statsmodels` and `pmdarima`, while machine-learning models are implemented using `scikit-learn`, including Random Forest and LASSO regressions. Visualization and diagnostic plots are generated using `matplotlib`.

A single entry point (`main.py`) orchestrates the full workflow, executing data preprocessing, model estimation under a rolling forecasting scheme, and final performance evaluation. Core functionality is organized into separate scripts corresponding to data loading, feature construction, model-specific forecasting pipelines, and evaluation utilities. Intermediate datasets and forecast outputs are persisted to disk at key stages of the pipeline to avoid unnecessary re-computation and facilitate controlled experimentation. All models operate on country-specific datasets, and no pooling across countries is performed at the estimation stage.

The data preprocessing layer constructs aligned country month panels from raw OECD and Trading Economics inputs, applies variable-specific transformations, and generates leakage-safe lagged feature matrices. Lag construction is centralized and applied consistently across models, ensuring that all predictors rely exclusively on information available at the forecast origin.

Model specific forecasting pipelines build on this common preprocessing layer. SARIMAX includes automated order selection and post-processing of CPI level forecasts into yoy inflation, while Random Forest and LASSO share identical feature matrices and forecasting loops to ensure comparability. For LASSO, standardization is implemented within a `Pipeline` object fitted on training data only, preventing information leakage by construction.

A dedicated script implements the high-dimensional Germany extension. This component

¹Appendix A.1 provides a visual illustration of the year-on-year log-difference transformation for Belgium, showing how the transformation removes deterministic seasonality present in CPI levels.

automatically discovers indicator files, infers valid numeric value columns, handles heterogeneous reporting frequencies, and applies indicator-specific transformations and lag structures, allowing the information set to be expanded with minimal changes to the core pipeline.

Final forecast evaluation is centralized in a comparison module that loads model specific forecast outputs and computes performance metrics using a uniform procedure.

```

1 def infer_value_column(df: pd.DataFrame, date_col: str) -> str:
2     """
3     Infer the numeric value column from a dataframe. Identify
4     dates that are numeric. Helps avoid treating dates as data
5     value
6
7     Parameters
8     -----
9     df : pd.DataFrame
10         Input dataframe.
11     date_col : str
12         Name of the date column to exclude.
13
14     Returns
15     -----
16     str
17         Name of the inferred value column.
18
19     Raises
20     -----
21     ValueError
22         If no suitable numeric column is found.
23     """
24
25     # Collect candidate numeric columns, excluding the known date column
26     numeric_cols = []
27
28     for col in df.columns:
29         if col == date_col:
30             continue
31
32         # Non-numeric entries become NaN
33         series = pd.to_numeric(df[col], errors="coerce")
34         valid_count = series.notna().sum()
35
36         # Minimum coverage threshold: avoid selecting sparse columns
37         # Rule is numeric value must cover either 30 percent of dataset or at
38         # least 5 observations
39
40         min_required = max(5, int(0.3 * len(df)))
41
42         if valid_count >= min_required:
43             # Store candidates as (column_name, number_of_valid_values)
44             numeric_cols.append((col, valid_count))
45
46     # Fail fast if no suitable numeric column exists
47     if not numeric_cols:
48         raise ValueError(
49             f"No numeric value column detected in {df.columns}"
50         )
51
52     # Column with the highest non-missing coverage
53     # is detected as the value we aim to use in the dataset
54     numeric_cols.sort(key=lambda x: x[1], reverse=True)
55     return numeric_cols[0][0]

```

Listing 1: Inference of the numeric value column, 5 prep data GER.py

4 Results

4.1 Experimental Setup

All experiments were executed locally on a personal computer using CPU computation. The system is equipped with an Intel(R) Core(TM) i5-6200U processor (2 physical cores, 4 threads, base frequency 2.30 GHz) and 16,GB of RAM, running a 64-bit Ubuntu Linux operating system (kernel version 5.15).

All experiments are executed through a single entry point (`main.py`) using Python version 3.11. The software environment is fully specified via a `requirements.txt` file, which records the versions of all external dependencies used to produce the results. This setup ensures that all reported results are fully reproducible across systems.

All hyperparameters are fixed prior to the 2019 out-of-sample period to avoid ex-post tuning, and models are evaluated under an expanding-window, one-step-ahead design.

SARIMAX). Seasonal ARIMA orders are selected using `pmdarima`'s `auto_arima` procedure, with monthly seasonality fixed at $m = 12$ and model selection based on the Bayesian Information Criterion (BIC). The search space is deliberately restricted to low-order dynamics, with non-seasonal parameters $p, q \in \{0, \dots, 3\}$ and $d \leq 2$, reflecting standard practice in macroeconomic time-series modeling where short-run persistence is typically captured by low-order ARMA components. Seasonal parameters are bounded to $P, Q \in \{0, \dots, 2\}$ to limit model complexity and avoid unstable estimation at long seasonal lags. Seasonal differencing D is determined endogenously by the procedure using seasonal unit root tests, following the default implementation.

Exogenous regressors enter the model exclusively in lagged form: for each country, lags of unemployment and import prices are pre-selected prior to evaluation and loaded from dedicated lag-selection outputs. SARIMAX models are estimated on CPI levels to capture seasonal dynamics directly, and level forecasts are subsequently transformed into year-on-year inflation for comparability with the machine-learning models.

Random Forest. The Random Forest is calibrated conservatively to balance flexibility and overfitting in a small-sample macroeconomic forecasting context. The ensemble consists of 500 trees to stabilize predictions through averaging, while tree depth is capped at 6 and the minimum leaf size is set to 15 observations, limiting local adaptivity and reducing sensitivity to short-lived noise or outliers in CPI inflation.

At each split, feature subsampling is applied with `max_features` set to one third of the available predictors, this lowers tree correlation while preserving predictive power in regression settings. Extensive hyperparameter tuning is deliberately avoided.

Feature construction is adapted time-series. Inflation enters with lags $[1, 3, 6, 12]$, capturing short-run persistence, quarterly dynamics, medium-term adjustment, and annual seasonality. Explanatory indicators (unemployment and import prices) enter with lags $[1, 2, 4]$, reflecting delayed transmission of labor-market and external price pressures. In addition, a 3-month inflation momentum feature is included to summarize short-horizon trends, improving forecast stability by conditioning on recent inflation direction rather than isolated monthly movements.

LASSO. LASSO regression is used as a disciplined linear benchmark, designed to perform automatic variable selection in a setting with potentially redundant lagged predictors. The model is implemented within a preprocessing **Pipeline** that standardizes all features using training data only, ensuring scale comparability across lagged variables while preventing look-ahead bias.

Feature construction mirrors Random Forest to maintain comparability across models. Inflation enters with lags $[1, 3, 6, 12]$, capturing short-run persistence, quarterly dynamics, and annual effects, while unemployment and import prices enter with short-horizon lags $[1, 2, 4]$, reflecting delayed but linear transmission channels. LASSO's ℓ_1 penalty then selects among these candidates, shrinking weak or redundant predictors toward zero.

The regularization parameter α is selected once per country using time-series cross-validation

(5 splits) on pre-test data over a fixed grid of 30 values, $\alpha \in \text{logspace}(-4, 1, 30)$. The selected value is held fixed throughout the 2019 evaluation window to avoid re-tuning on test data. Convergence is ensured via conservative optimizer settings (`max_iter` = 20000, `tol` = 10^{-4}).

4.2 Performance Evaluation

Forecast accuracy is assessed using two standard loss functions for continuous outcomes: the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE). RMSE penalizes large forecast errors more heavily and is therefore sensitive to occasional large deviations, while MAE provides a linear measure of average forecast error that is more robust to outliers. Using both metrics provides a balanced assessment of predictive performance.

All models produce forecasts of year-on-year CPI inflation. When estimation is conducted on different internal representations, such as CPI levels for SARIMAX, forecasts are transformed to a common YoY inflation scale prior to evaluation. Performance metrics are computed on matched forecast–realization pairs over the common out-of-sample period to ensure comparability across models.

Forecast errors are first aggregated at the country level by computing RMSE and MAE over the 12-month evaluation window. These country-specific metrics are then compiled into summary tables that facilitate both within-country model comparisons and cross-country performance assessments.

All evaluation steps are implemented in a dedicated comparison module, which loads model-specific forecast outputs from disk and computes RMSE and MAE using a uniform procedure. Centralizing performance evaluation ensures that all reported results are based on identical metrics, aggregation rules, and evaluation windows.

Table 1: Cross-country RMSE comparison

Country	SARIMAX	Random Forest	LASSO	Best model
Belgium	0.129	0.107	0.149	Random Forest
Germany	0.251	0.212	0.317	Random Forest
Israel	0.243	0.188	0.243	Random Forest
Korea	0.156	0.604	0.195	SARIMAX
Latvia	0.335	1.123	0.372	SARIMAX
Lithuania	0.147	0.142	0.147	Random Forest
Norway	0.187	0.182	0.164	LASSO
Switzerland	0.239	0.226	0.197	LASSO

Table 2: Germany: RMSE comparison across model specifications

Country	SARIMAX	LASSO	RF (baseline)	RF (high-dimensional)
Germany	0.251	0.317	0.212	0.227

Table 1 reports root mean squared error (RMSE) values for year-on-year CPI inflation forecasts across eight countries and all three models. The table highlights substantial heterogeneity in forecasting performance across countries. Random Forest achieves the lowest RMSE in four countries (Belgium, Germany, Israel, and Lithuania), while SARIMAX performs best in Korea and Latvia. LASSO attains the lowest RMSE in Norway and Switzerland. Across all model classes, Latvia consistently exhibits the largest forecast errors. Table 2 focuses on Germany and compares alternative model specifications. The Random Forest with two feature set yields the lowest RMSE among all models, followed by SARIMAX. The high-dimensional Random Forest

specification does not outperform the lower-dimensional Random Forest. This table provides a focused comparison for Germany and motivates further analysis of feature dimensionality in Random Forest implementation.

4.3 Visualizations

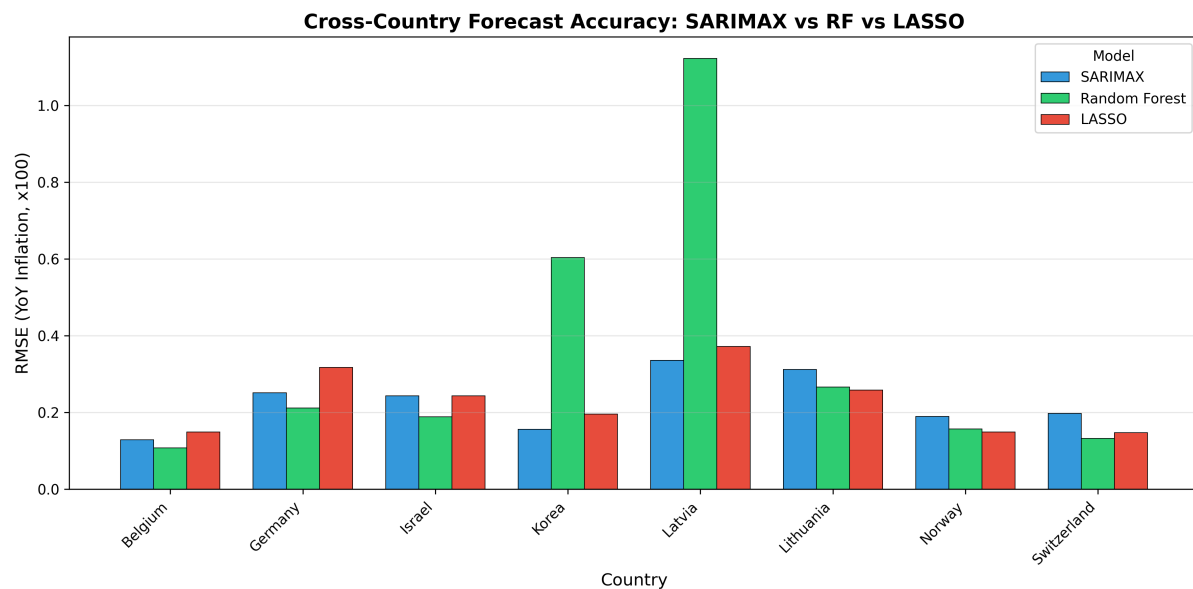


Figure 1: Cross-country forecast accuracy comparison across models.

Figure 1 provides a visual comparison of forecast accuracy across countries and models. Random Forest performance varies substantially across countries, with notably higher forecast errors for Latvia and Korea relative to other economies. In contrast, differences between models are smaller for countries such as Belgium, Israel, and Lithuania, where all three approaches yield relatively similar error magnitudes. Figure 2 presents the Random Forest forecast for

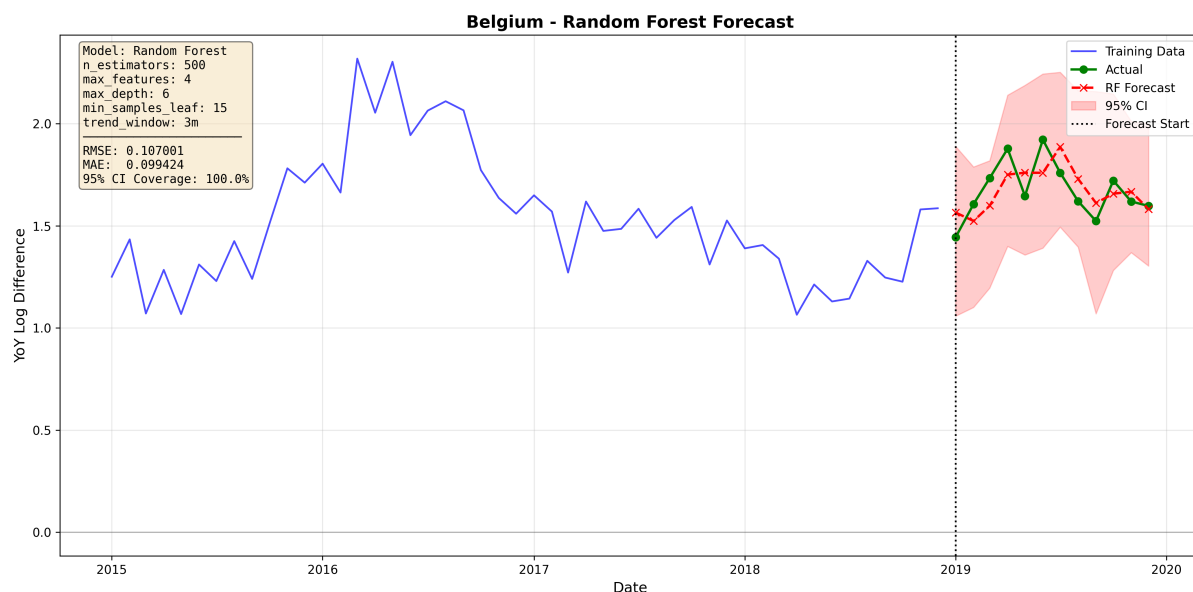


Figure 2: Random Forest forecast for Belgium.

Belgium (lowest RMSE) over the out-of-sample period. The figure shows the training data and

the one-step-ahead forecasts.

5 Discussion

5.1 Overall forecasting performance

All three approaches : SARIMAX, Random Forest and LASSO deliver reasonably accurate forecasts of yoy CPI inflation. Forecast errors remain relatively low for most countries, suggesting that a substantial share of short run inflation dynamics can be captured using a limited set of lagged variables and a small number of exogenous indicators. This highlights the difficulty of achieving systematic improvements over simple benchmark dynamics: (Beck, Kozbur, and Wolf 2024) document that inflation forecasts are notoriously hard to improve beyond univariate benchmark models, such as random walk type specifications, highlighting the dominant role of persistence at short horizons. Moreover one-step-ahead predictive performance is often maximized using a small number of recent lagged predictors, reinforcing the importance of short run dependence and limited memory in forecasting settings (Tyralis and Papacharalampous 2017).

5.2 Model-specific performance patterns

The LASSO model performs particularly well for Switzerland and Norway. These two economies are characterized by relatively stable inflation dynamics and limited short-run volatility over the sample period. In such environments, shrinkage-based linear models tend to concentrate predictive weight on recent inflation realizations while shrinking weak or noisy predictors toward zero, yielding forecasts that behave similarly to highly persistent benchmark dynamics. In the context of inflation forecasting, Beck and Wolf emphasize that achieving consistent improvements over simple univariate benchmarks remains notoriously difficult, which is consistent with the idea that, in low-volatility settings, parsimonious and strongly regularized specifications can be competitive (Beck, Kozbur, and Wolf 2024).

SARIMAX models also perform consistently well across countries. Their advantage lies in the explicit modeling of autoregressive and seasonal components, which allows them to reproduce recurring patterns observed in monthly CPI data. The fact that SARIMAX outperforms its non-seasonal ARIMAX counterpart in this project highlights the empirical relevance of seasonality when forecasting monthly inflation dynamics.

A notable result is the strong overall performance of the Random Forest model, which matches or exceeds SARIMAX accuracy in several countries despite relying on a relatively small feature set. Beck and Wolf motivate the use of machine learning for inflation forecasting by arguing that machine learning methods deliver their most robust gains when they are applied to rich information sets (i.e., a large number of features), and they review evidence that such high-dimensional approaches can outperform traditional benchmark models (Beck, Kozbur, and Wolf 2024) In this project, Random Forest performance remains competitive even with a moderate feature set, which is consistent with evidence that one-step-ahead Random Forest forecasting can be maximized using a low number of recent lagged predictors when the feature design is aligned with the time-series structure (Tyralis and Papacharalampous 2017).

5.3 Latvia, Korea, and Random Forest underperformance

Latvia exhibits the highest forecast errors across all models. During data preprocessing, Latvia also displayed higher volatility and a shorter effective training sample after transformations, as unemployment and import price series begin later than for other countries. A reduced sample length limits the information available for model estimation and likely contributes to weaker overall forecasting performance.

In addition to Latvia, the Random Forest model performs particularly poorly for Korea ², while LASSO and SARIMAX do not display a comparable deterioration. This suggests that the issue is not driven by the underlying data, but is instead specific to the Random Forest specification. Visual inspection of the Random Forest forecasts indicates that predicted dynamics broadly track realized inflation paths pattern for both countries, but appear systematically shifted in levels. One possible explanation is a scaling or normalization issue; however, extensive code checks and the uniform looping structure applied across all countries make this explanation difficult to verify.

An alternative explanation relates to the timing of inflation dynamics at the end of the training sample. For both Latvia and Korea, inflation rises sharply toward the end of 2018, immediately preceding the out-of-sample period. Because this shift occurs very late in the training window, the Random Forest may not have sufficient observations to learn and generalize this behavior, resulting in biased level forecasts. The absence of a similar issue for LASSO and SARIMAX suggests that models with stronger parametric structure or explicit persistence mechanisms may be better suited to handle late sample level shifts.

More broadly, these findings are consistent with evidence that inflation predictability varies substantially across countries, particularly for smaller and more open economies exposed to idiosyncratic shocks (Stock and Watson 2016). The precise source of Random Forest underperformance in Latvia and Korea, however, cannot be conclusively identified and remains an important limitation of the analysis.

5.4 High-dimensional extension for Germany

An important negative result concerns the high-dimensional Random Forest extension for Germany. Contrary to expectations, augmenting the baseline feature set with additional macroeconomic indicators leads to a slight deterioration in forecast accuracy. One explanation is that increasing the number of predictors without a proportional increase in sample size raises estimation variance and leads to less reliable split decisions in tree-based models (Dudek 2015). In addition, many macroeconomic indicators are highly correlated, which reduces the effective information gain from additional features and weakens the contribution of individual splits (Dudek 2015).

A further explanation relates to the information content of monthly inflation data at short horizons. Beck and Wolf emphasize that improvements from Random Forest-based approaches in inflation forecasting depend critically on the availability of sufficiently informative predictors, and that short-horizon monthly inflation dynamics may offer limited exploitable signal beyond a parsimonious set of variables (`beck_wolf_rf`). In this context, expanding the feature space can introduce noise rather than additional predictive content.

This result highlights the importance of aligning model complexity with data availability, forecast horizon, and information content.

5.5 Limitations

First, the evaluation focuses on a short forecast horizon and excludes the COVID-19 period, which limits the applicability of the results to episodes of extreme structural change and regime shifts in inflation dynamics.

Second, the baseline comparison relies on a highly parsimonious set of exogenous variables. This choice is partly dictated by econometric considerations: in macroeconomic applications, identifying truly exogenous variables is difficult, as most macro indicators are strongly correlated with inflation, while variables that are plausibly exogenous often exhibit limited predictive power. As a result, the SARIMAX specification necessarily relies on a small information set, which

²See Appendix A.2 for forecast plots of Latvia and Korea.

constrains its ability to exploit richer macroeconomic signals. In contrast, machine-learning models such as Random Forests can accommodate large and highly correlated predictor sets, but this flexibility does not guarantee improved forecasting performance.

Third, the Germany high-dimensional extension shows that expanding the feature space without simultaneously adapting model design and tuning is insufficient. Simply adding indicators does not automatically improve forecast accuracy and may introduce additional noise, particularly in short-horizon settings. The performance of Random Forest inflation forecasts depends on careful model specification and hyperparameter calibration when the information set is expanded (Beck, Kozbur, and Wolf 2024). In this project, conservative hyperparameter choices proved more robust, suggesting that gains from richer feature sets require more deliberate tuning strategies rather than naïve feature augmentation.

6 Conclusion

6.1 Summary

Random Forest models achieve the strongest overall forecasting performance, outperforming econometric benchmarks in several countries, though with notable underperformance in two cases. In contrast, SARIMAX delivers more stable performance across countries, reflecting the importance of explicitly modeling persistence and seasonality in monthly inflation data. These results indicate that carefully designed machine-learning models can compete with traditional time-series approaches even in moderate dimensional settings. At the same time, the findings highlight that increasing model complexity or feature dimensionality does not automatically translate into improved forecast accuracy. Overall, the results suggest that machine learning methods can deliver competitive short-horizon inflation forecasts in small-sample settings, but do not systematically dominate well-specified econometric benchmarks.

6.2 Future Directions

Future work could focus on improving Random Forest performance. First, introducing a simple variable-selection step within each training window, such as **retaining only the most important predictors** based on permutation importance, which could reduce noise and variance. Second, additional indicators could be incorporated through factor compression, for example by extracting a small number of principal components from the predictor set, which would allow richer information to be included while controlling dimensionality. Third, further restricting lag depth for auxiliary indicators while preserving richer dynamics for inflation itself may improve short-horizon performance. Finally, adapting key Random Forest hyperparameters to the dimensionality of the feature space, or implementing weighted ensemble variants such as hedged Random Forests, represents a promising direction for improving robustness in inflation forecasting applications.

References

References

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Code Resources

The following online resources were consulted for the code part. These sources are not part of the academic literature review.

- <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html> (accessed January 2026)
- <https://labex.io/fr/tutorials/python-how-to-train-random-forest-in-scikit-learn-425422> (accessed January 2026)
- <https://www.geeksforgeeks.org/dsa/random-forest-classifier-using-scikit-learn/> (accessed January 2026)
- <https://data36.com/random-forest-in-python/> (accessed January 2026)
- <https://stackoverflow.com/> (accessed January 2026)
- <https://stats.stackexchange.com/> (accessed January 2026)
- <https://towardsdatascience.com/time-series-forecasting-with-arima-sarima-and-sarimax-ee61099e78f6/> (accessed January 2026)
- https://skforecast.org/0.10.0/user_guides/forecasting-sarimax-arima (accessed January 2026)
- https://www.sktime.net/en/latest/api_reference/auto_generated/sktime.forecasting.arima.AutoARIMA.html (accessed January 2026)

A Additional figures

A.1 Year-on-Year Log-Difference Transformation

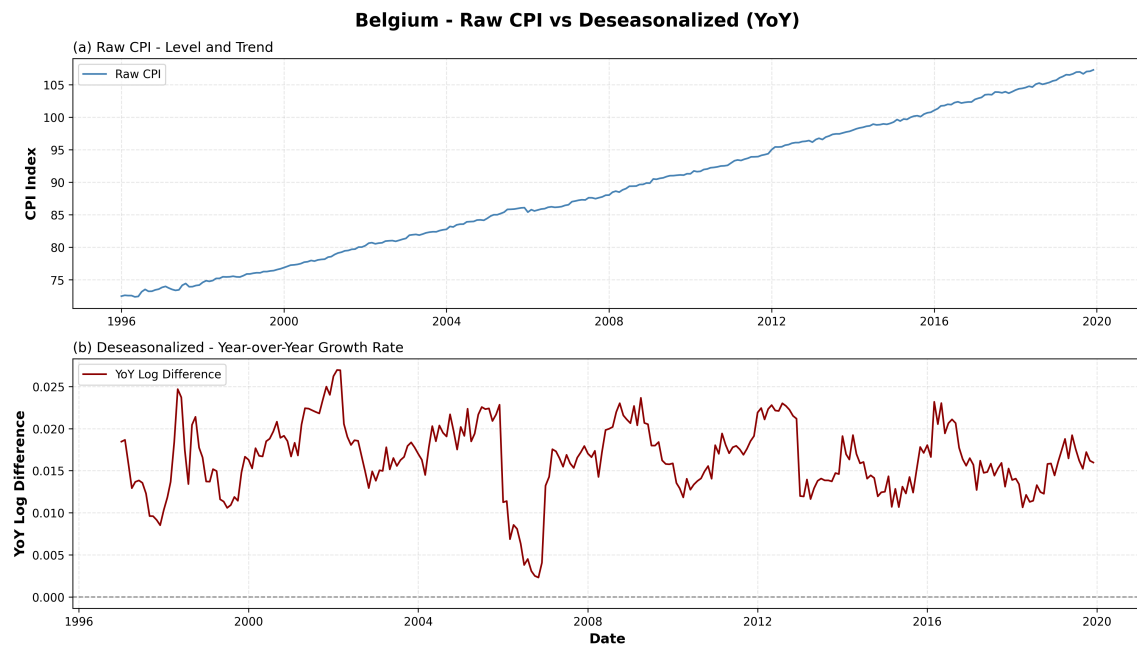


Figure 3: Raw CPI levels and year-on-year log-difference transformation for Belgium.

Figure 3 contrasts raw CPI levels with the corresponding year-on-year log-difference series for Belgium. While CPI levels exhibit a strong trend and seasonal variation, the year-on-year transformation removes deterministic seasonality and stabilizes the series around a relatively constant mean.

A.2 Additional Forecast Diagnostics

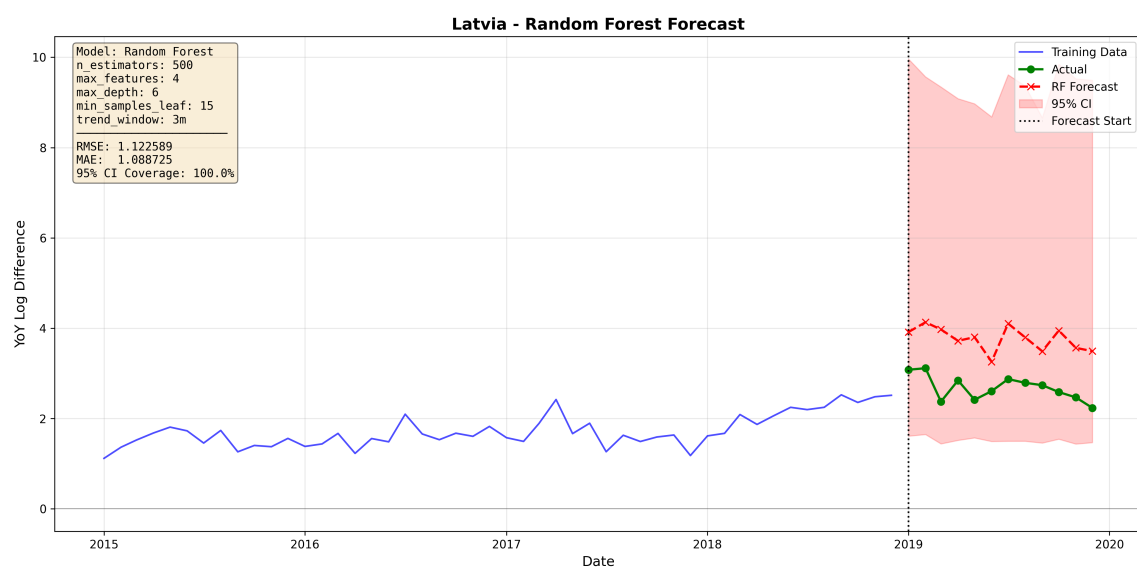


Figure 4: Random Forest one-step-ahead forecast for Latvia.

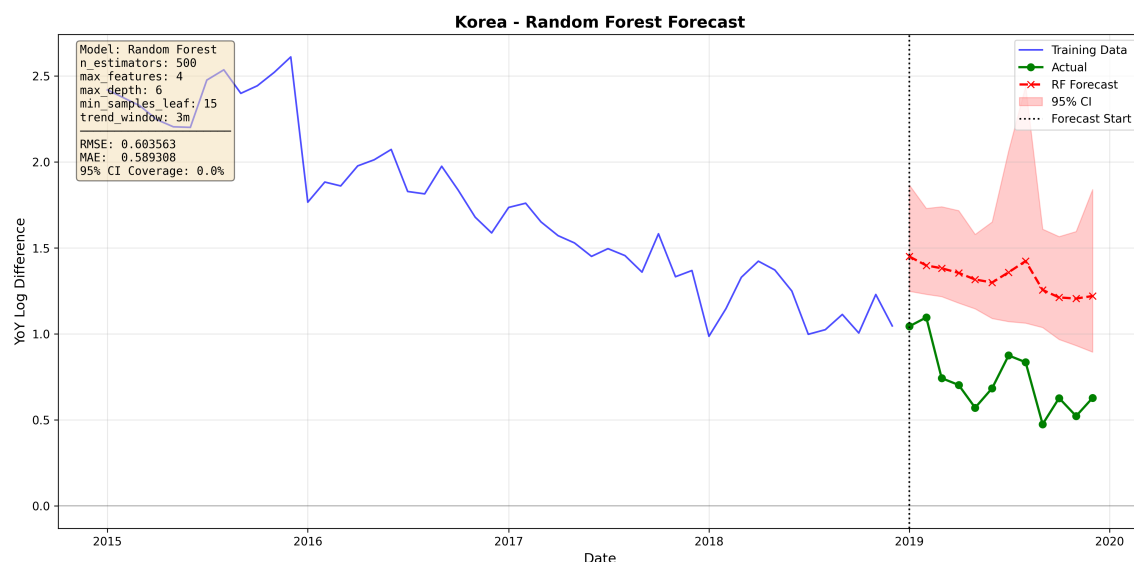


Figure 5: Random Forest one-step-ahead forecast for Korea.

Figures 4 and 5 illustrate Random Forest forecasts for Latvia and Korea, respectively. In both cases, the forecast paths closely follow the direction and short-term dynamics of realized year-on-year inflation, indicating that the model captures temporal patterns reasonably well. However, forecasts appear systematically shifted upward relative to realized values, suggesting a scaling or level calibration issue rather than a failure to learn underlying dynamics.

A.3 Benchmark Forecasts for Belgium

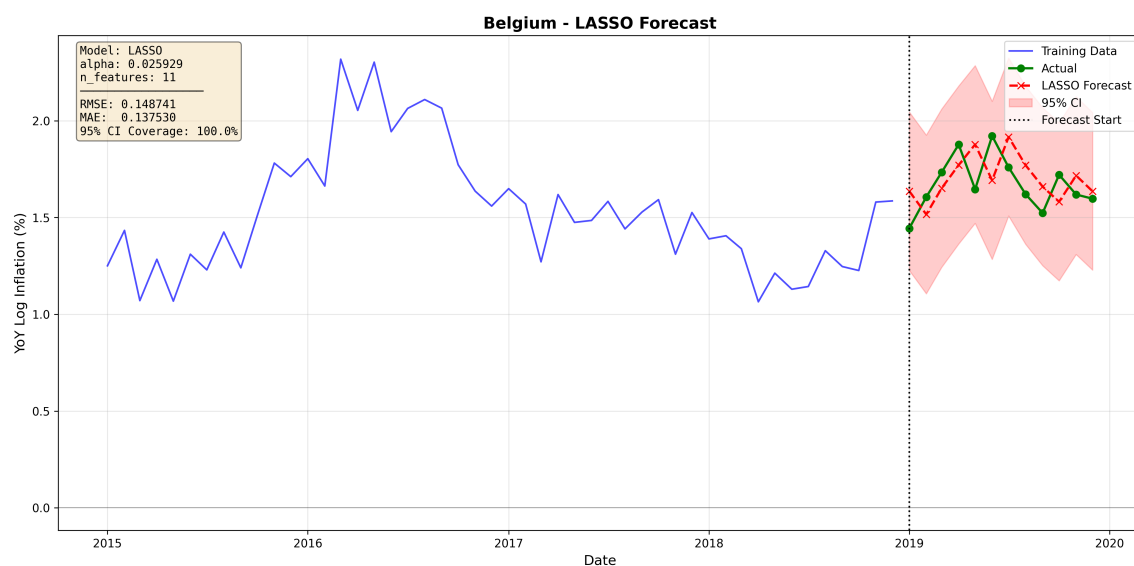


Figure 6: LASSO one-step-ahead forecast for Belgium.

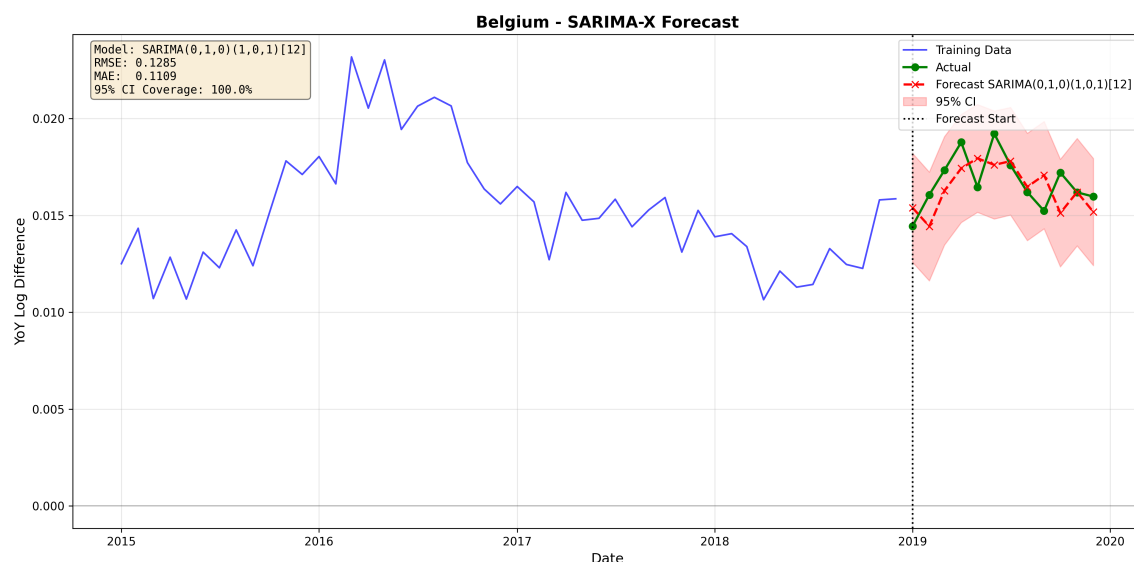


Figure 7: SARIMAX one-step-ahead forecast for Belgium.

Figures 6 and 7 present one-step-ahead forecasts for Belgium, which serves as a representative benchmark case. For this country, both LASSO and SARIMAX closely track observed inflation dynamics over the evaluation period, consistent with their strong quantitative performance.

B Code Repository

Use of AI

Artificial intelligence tools were used in a limited and supportive role during the preparation of this project. AI tools were used for the following purposes:

- Formatting and organizing Python code to ensure compliance with project requirements and coding standards (PEP-8).
- Assisting with debugging by helping identify syntax errors, implementation issues, and potential improvements in code structure.
- Supporting brainstorming and design decisions during code development.
- Assisting with project organization and submission readiness by cross-checking required components and identifying potential gaps.
- Improving clarity, structure, and wording of the written report.

All data processing, model implementation, empirical analysis, interpretation of results, and final conclusions were designed, implemented, and validated by the author. AI tools were not used to generate results, select models, or influence empirical findings.

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GitHub Repository: <https://github.com/eonieva24/Macroeconomic-Forecasting>